Gaussian Eliminating with Partial Pivoting

$$\begin{pmatrix} 2 & 5 & 8 \\ 4 & 6 & 2 \\ 7 & 2 & 6 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4 \\ 8 \\ 9 \end{pmatrix}$$

First check A matrix large value

In this matrix 7 is large.

then Interchange the row

$$\begin{pmatrix} 7 & 2 & 6 \\ 4 & 6 & 2 \\ 2 & 5 & 8 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 9 \\ 8 \\ 4 \end{pmatrix}$$

then we do the gaustan Elimination.

do the row operation & then Find the x, y, z values.

$$\left(\begin{array}{ccc|c} 7 & 2 & 6 & 9 \\ 4 & 6 & 2 & 8 \\ 2 & 5 & 8 & 4 \end{array}\right) \rightarrow (A \mid b)$$

$= 3(12-40)-2(10-28)$

$-8 \quad 4+124+99. \qquad +1(25+54)$

$$\left(\begin{array}{ccc|c} 7 & 2 & 6 & 9 \\ 4 & 6 & 2 & 8 \\ 2 & 5 & 8 & 4 \end{array}\right)$$

$R_2 - \frac{4}{7} R_1 \to R_2$

4	6	2	8
4	$\frac{8}{7}$	$\frac{24}{7}$	$\frac{36}{7}$
0	$\frac{34}{7}$	$\frac{-10}{7}$	$\frac{20}{7}$

$$\left(\begin{array}{cccc} 7 & 2 & 6 & 9 \\ 0 & \frac{34}{7} & \frac{-10}{7} & \frac{20}{7} \\ 2 & 5 & 8 & 4 \end{array}\right)$$

$R_3 - \frac{2}{7} R_1 \to R_3$

2	5	8	4
2	$\frac{4}{7}$	$\frac{12}{7}$	$\frac{18}{7}$
0	$\frac{31}{7}$	$\frac{44}{7}$	$\frac{10}{7}$

$$\left(\begin{array}{cccc} 7 & 2 & 6 & 9 \\ 0 & \frac{34}{7} & \frac{-10}{7} & \frac{20}{7} \\ 0 & \frac{31}{7} & \frac{44}{7} & \frac{10}{7} \end{array}\right)$$

$R_3 - \frac{31}{34} R_2 \to R_3$

0	$\frac{31}{7}$	$\frac{44}{7}$	$\frac{10}{7}$
0	$\frac{31}{7}$	$\frac{-155}{119}$	$\frac{310}{119}$
0	0	$\frac{129}{17}$	$\frac{-20}{17}$

$$\left(\begin{array}{cccc} 7 & 2 & 6 & 9 \\ 0 & \frac{34}{7} & \frac{-10}{7} & \frac{20}{7} \\ 0 & 0 & \frac{129}{17} & \frac{-20}{17} \end{array}\right)$$

→ use back substitution

$$x = \begin{pmatrix} 1.2636 \\ 0.5426 \\ -0.1550 \end{pmatrix}.$$